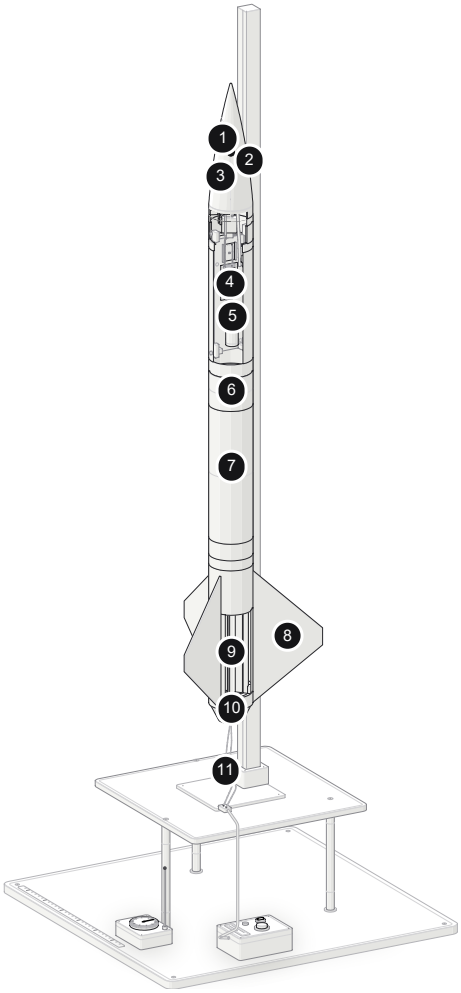
Helios: how a 38 mm rocket flies

HX-001



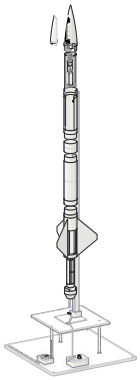
PARTS

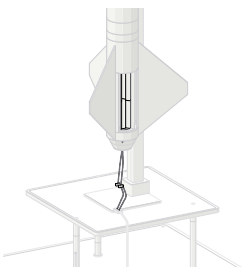
01 Side-door nose
02 Door
03 Parachute
04 Avionics bay
05 Main body
06 Coupler sleeve
07 Lower body
08 Fin can
09 38 mm KNSB motor
10 Boattail motor cap
11 Launch pad and rail

SPECIFICATION

Airframe	65.5 mm, five printed sections
Motor	38 mm KNSB, 2 BATES grains
Burn	1.3 s, 218 N·s, 170 N peak
Rail exit	16.6 m/s off 1.2 m
Apogee	about 800 m at T+12 s
Recovery	servo side door, 0.76 m canopy
Descent	6.5 m/s

TAKEN APART





01
On the rail



02
The grains burn



03
The barometer watches



04
The door kicks off



05
Under the canopy